V4.6 Comprehensive Statistical Summary

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CAR V4.6 INTRINSIC MODEL AUDIT | car_model_report_v4_6
================================================================
WAIC (Deviance): 4257.91 | LOO-CV: 4413.02
Divergences: 0 | Max R-hat: 1.910

V4.6 SPECIFICATION:
- Fixed Alpha: 0.99 (Intrinsic CAR)
- Centered Interaction: bsmokecentered * (bmen - 0.5)
- Variance Anchor: HalfNormal(0.5)

PUBLIC HEALTH INTERPRETATION:
- Pop. Smoking Effect: +2.02% risk per 5.3% increase
- Gender Modifier (Men): +0.10% additional risk adjustment
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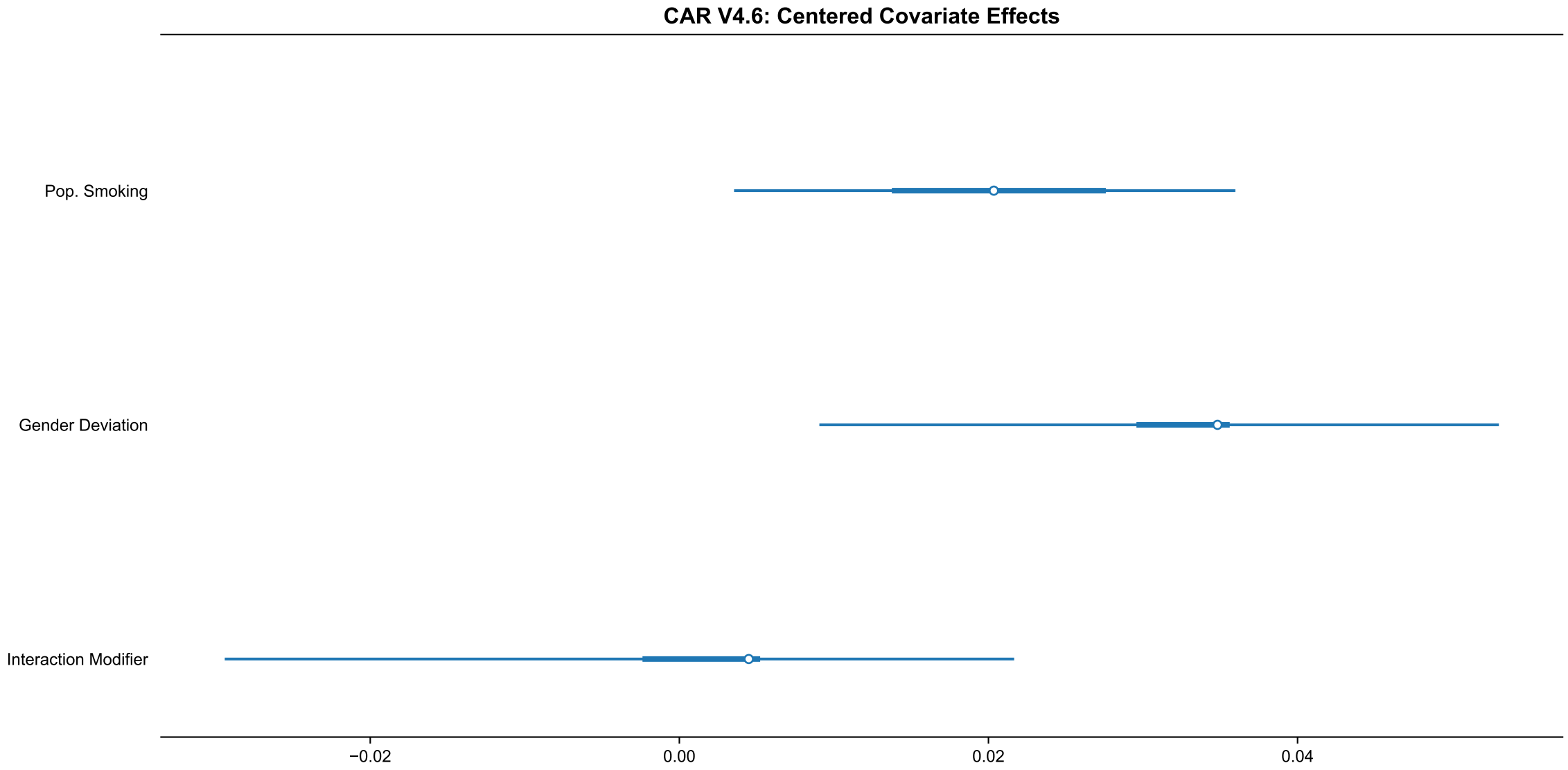
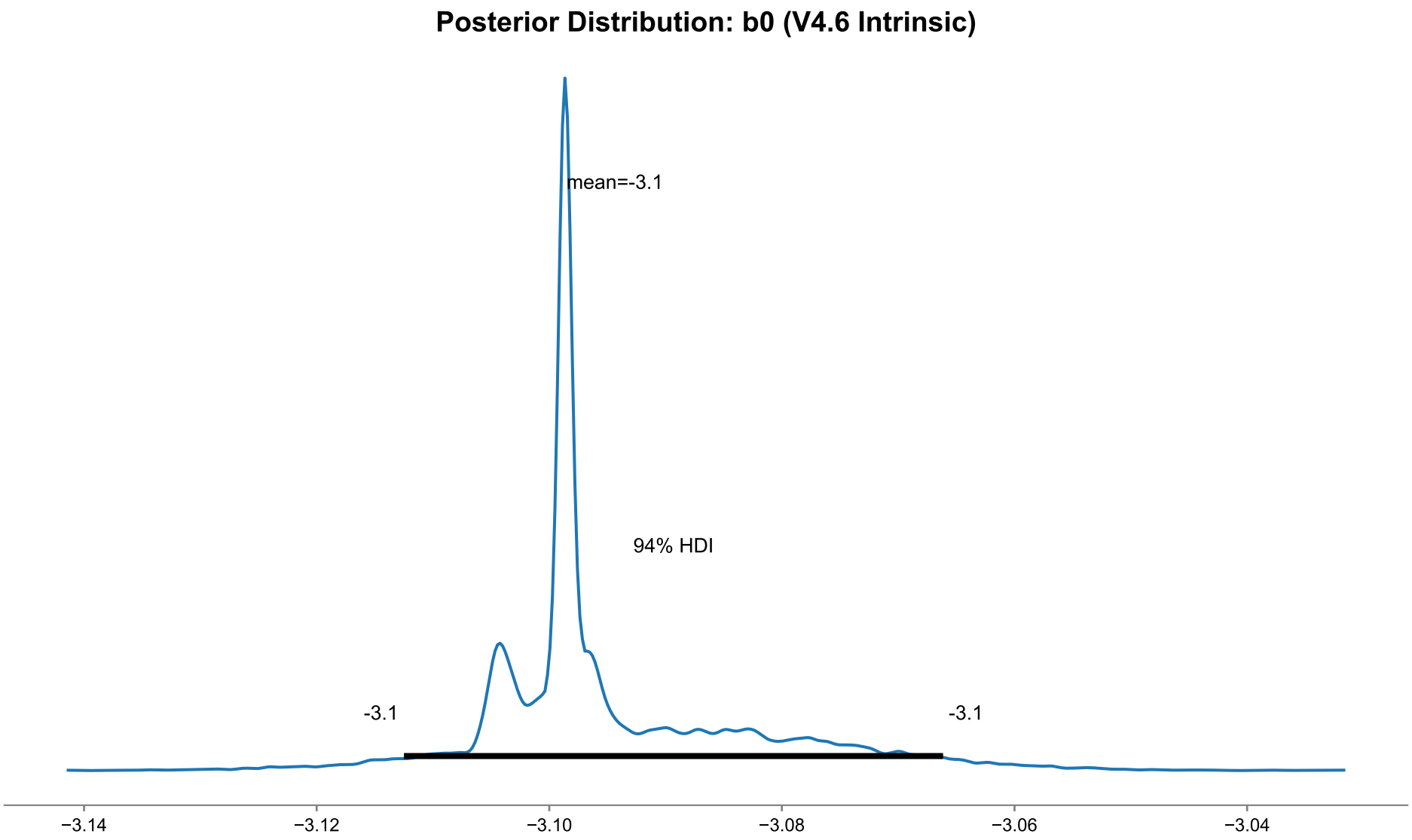
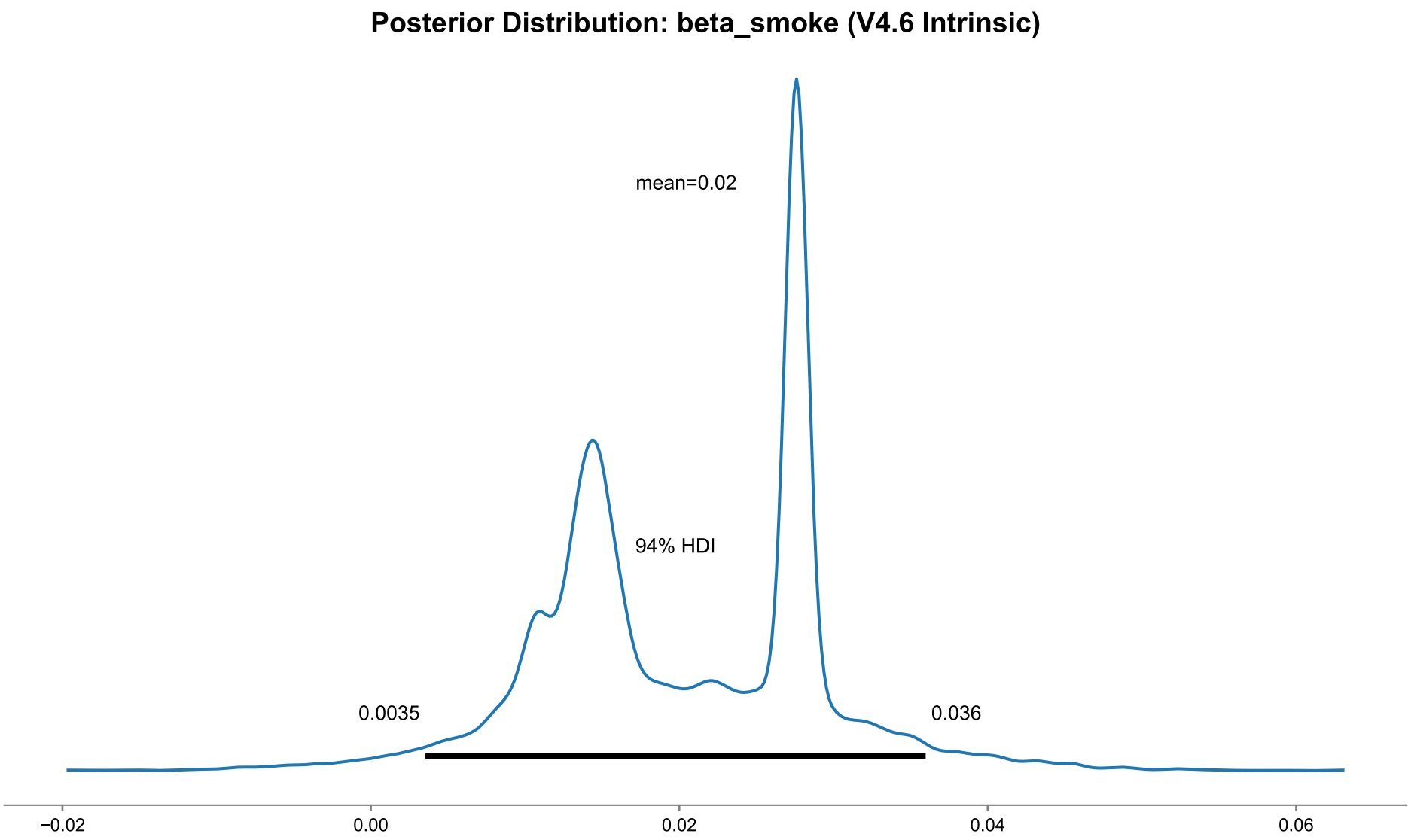
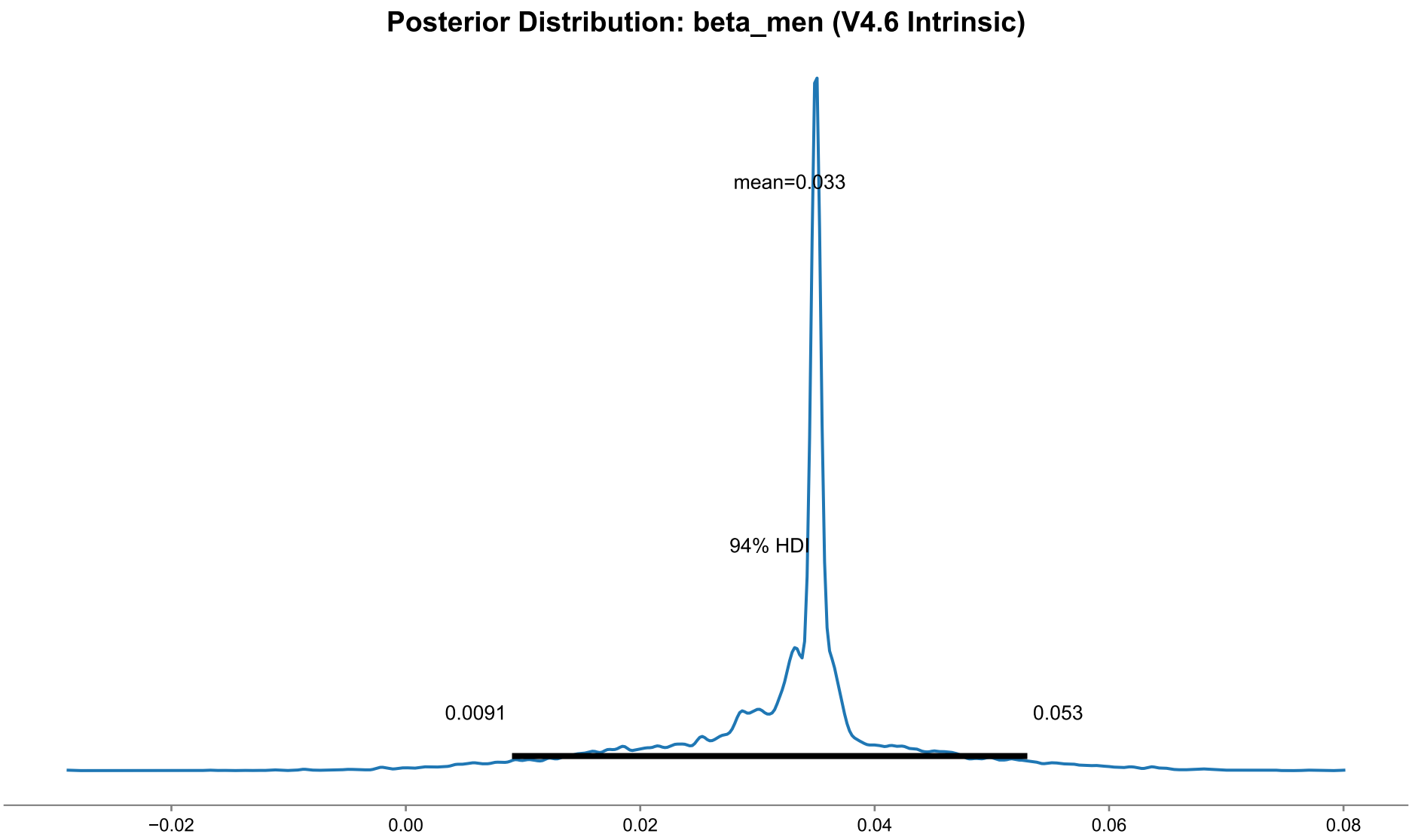
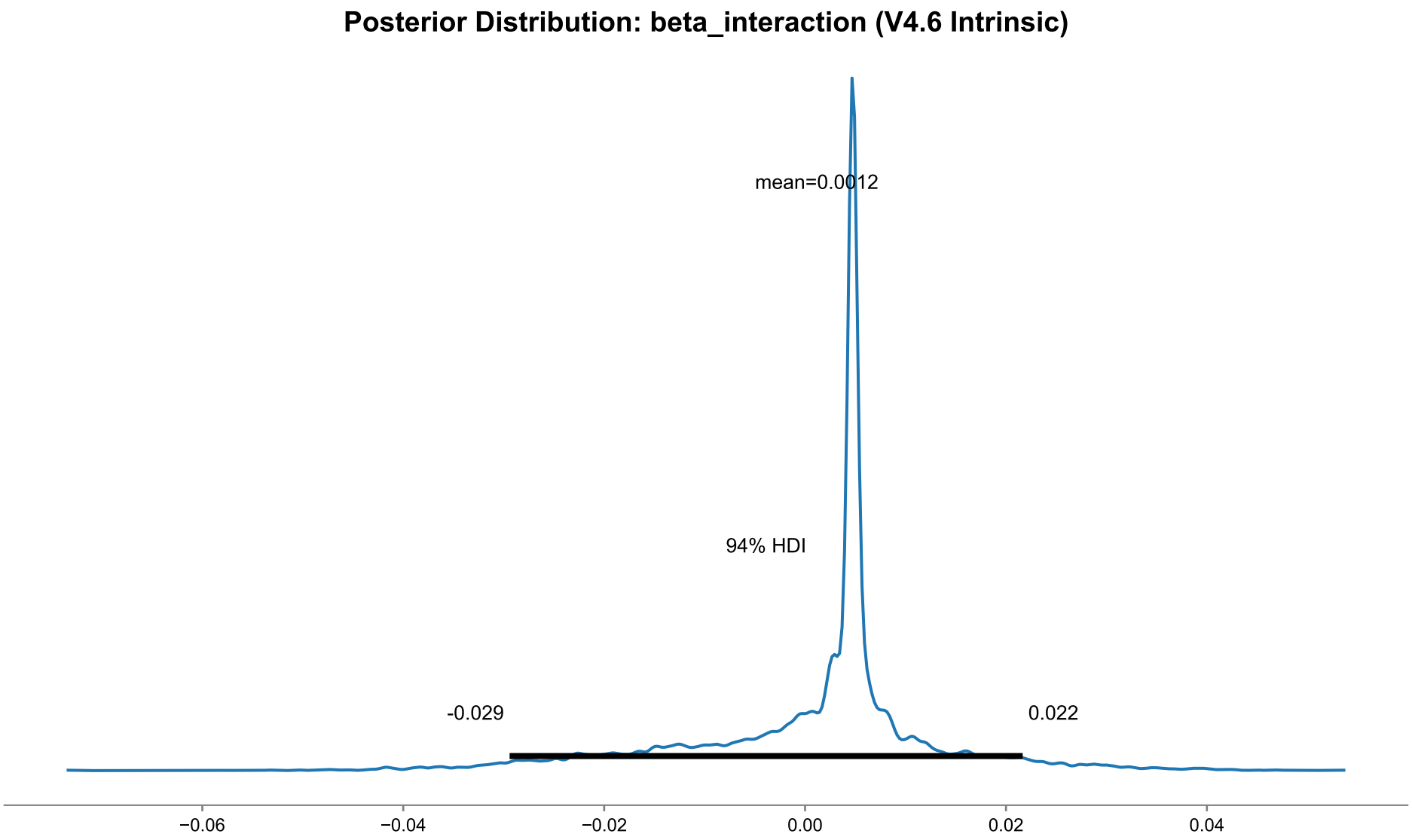
	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
b0	-3.094	0.012	-3.112	-3.066	0.003	0.002	17.0	69.0	1.91
beta_smoke	0.020	0.009	0.004	0.036	0.003	0.001	14.0	91.0	1.19
beta_men	0.033	0.010	0.009	0.053	0.001	0.002	261.0	97.0	1.71
beta_interaction	0.001	0.012	-0.029	0.022	0.002	0.003	61.0	95.0	1.77
sigma	0.257	0.025	0.216	0.297	0.009	0.004	8.0	49.0	1.70
rho	0.208	0.133	0.000	0.342	0.040	0.023	12.0	34.0	1.72

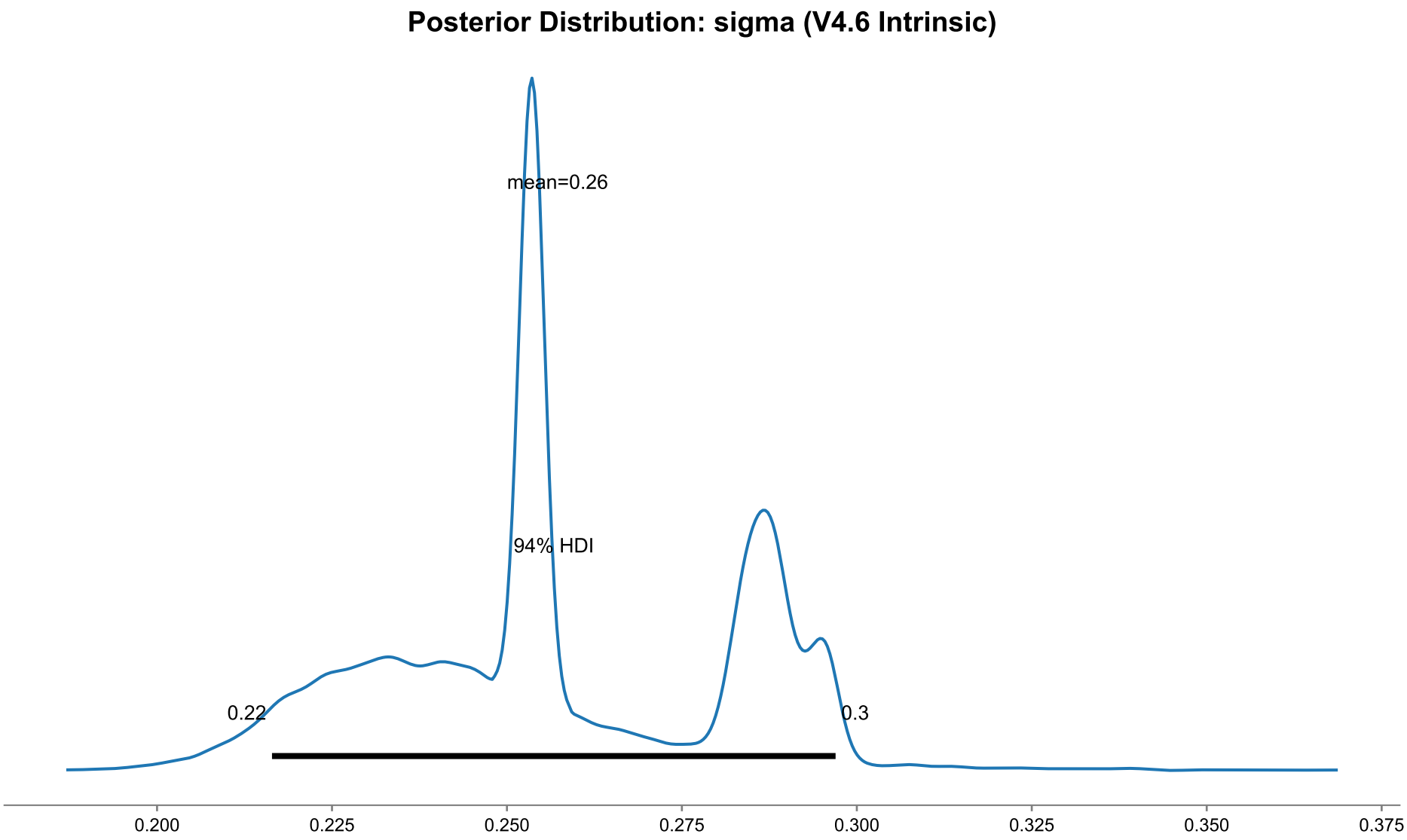

CAR V4.6: Centered Covariate Effects
Pop. Smoking
Gender Deviation
Interaction Modifier
−0.02
0.00
0.02
0.04

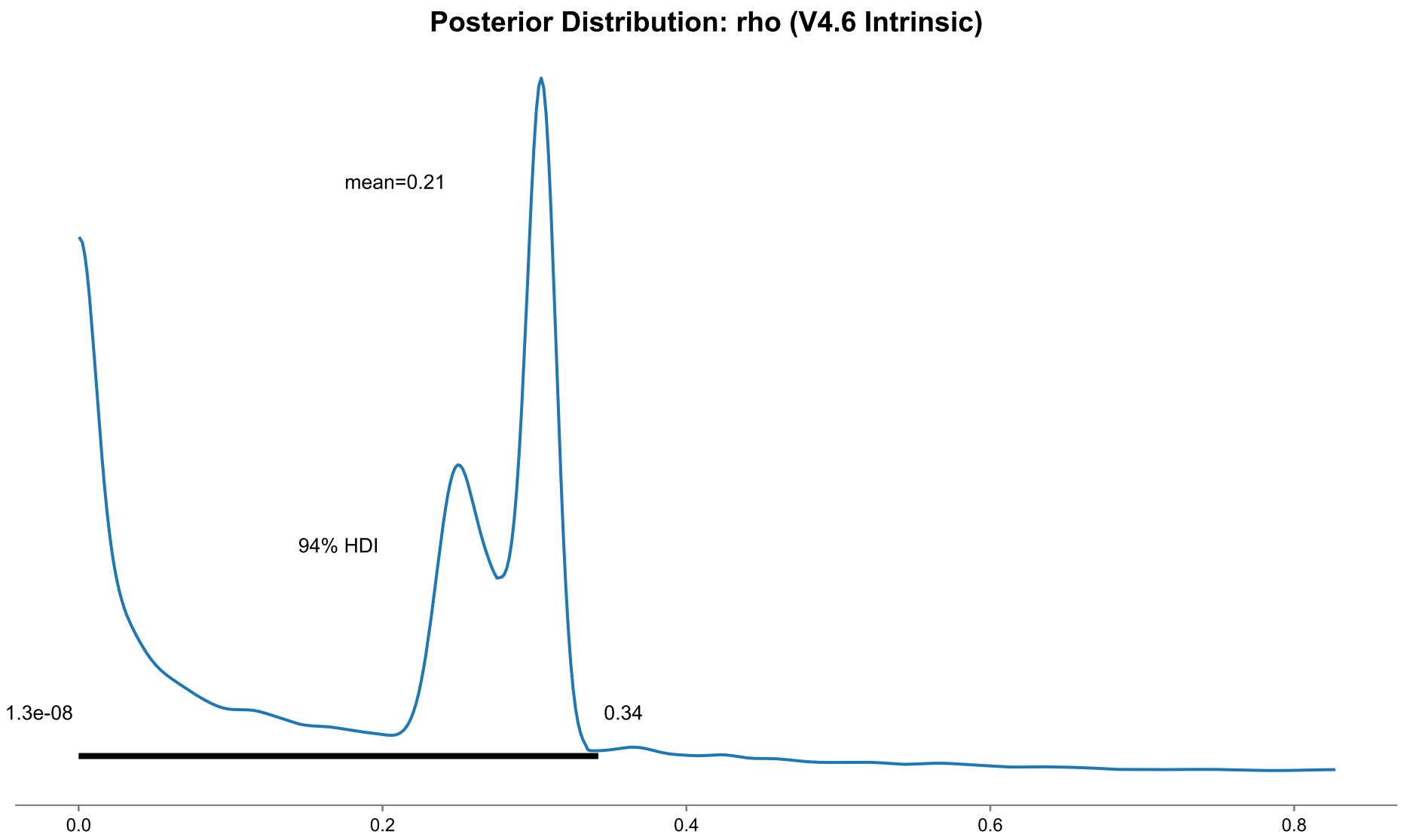
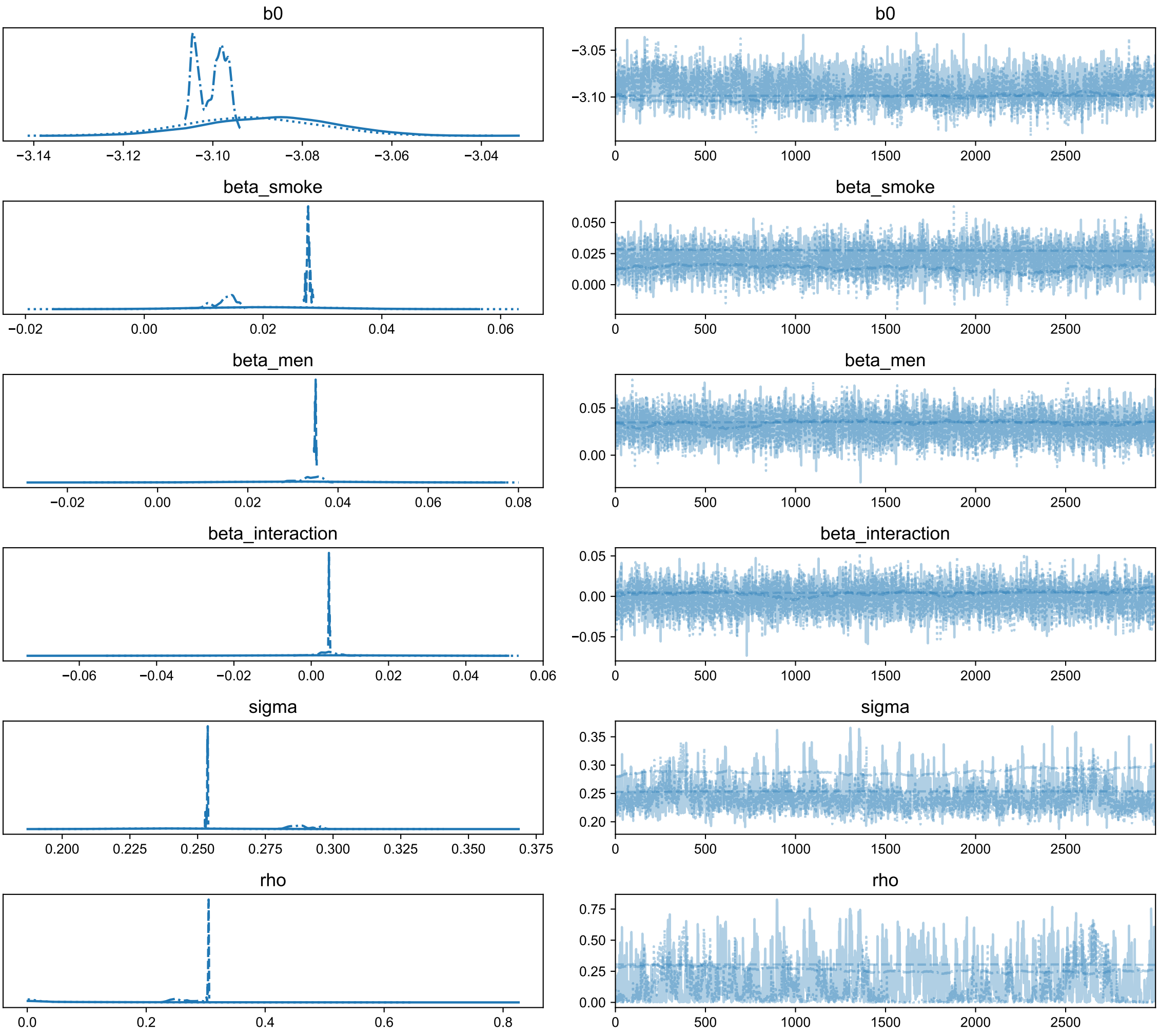

Posterior Distribution: b0 (V4.6 Intrinsic)
mean=-3.1
94% HDI
-3.1
-3.1
−3.14
−3.12
−3.10
−3.08
−3.06
−3.04


Posterior Distribution: beta_smoke (V4.6 Intrinsic)
mean=0.02
94% HDI
0.0035
0.036
−0.02
0.00
0.02
0.04
0.06


Posterior Distribution: beta_men (V4.6 Intrinsic)
mean=0.033
94% HDI
0.0091
0.053
−0.02
0.00
0.02
0.04
0.06
0.08


Posterior Distribution: beta_interaction (V4.6 Intrinsic)
mean=0.0012
94% HDI
-0.029
0.022
−0.06
−0.04
−0.02
0.00
0.02
0.04


Posterior Distribution: sigma (V4.6 Intrinsic)
mean=0.26
94% HDI
0.22
0.3
0.200
0.225
0.250
0.275
0.300
0.325
0.350
0.375


Posterior Distribution: rho (V4.6 Intrinsic)
mean=0.21
94% HDI
1.3e-08
0.34
0.0
0.2
0.4
0.6
0.8

V4.6 MCMC Chain Traces